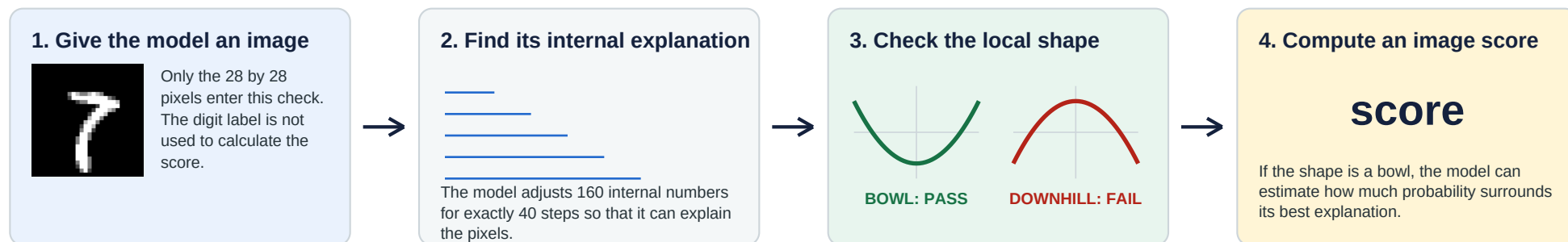


What this experiment tested



The goal

The full experiment asks whether independently trained image models can decide which temporal node should handle a new image. A node that learned images like the query should assign it a higher image-probability score. This could provide routing without a context label.

The 64-image safety check

Before building the large temporal bank, one trained model was tested on 64 unseen MNIST images. The score had to be computable for at least 99 percent of them. With only 64 images, that requirement means all 64 had to pass: 63 of 64 would be only 98.44 percent.

The allowed correction

The test tried three tiny, globally shared corrections: 0.00000001, 0.000001, and 0.0001. The same correction would have to work for every node and every image.

60 of 64 passed

The required result was 64 of 64.

Four images ended with one downhill direction. The probability approximation assumes a bowl, so it was not valid for those four images. The largest allowed correction was far too small to turn any of the four downhill shapes into bowls.

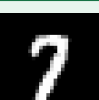
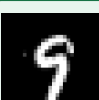
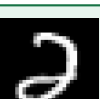
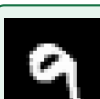
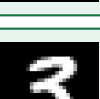
What happened next

The program stopped exactly where the protocol said it should stop. It did not build the temporal model bank. It did not test routing. It did not test merging. This result therefore says nothing yet about whether the proposed routing method works.

The exact 64 images

Green cells produced a valid score. Red cells did not. Each cell shows audit number, true digit, and predicted digit

60 pass | 4 fail | gate requires 64 pass

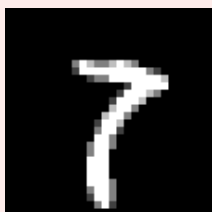
 #00 PASS true 2 pred 2	 #01 PASS true 4 pred 4	 #02 PASS true 4 pred 7	 #03 PASS true 0 pred 0	 #04 PASS true 8 pred 8	 #05 PASS true 7 pred 7	 #06 PASS true 7 pred 7	 #07 PASS true 0 pred 0
 #08 PASS true 3 pred 3	 #09 PASS true 5 pred 5	 #10 PASS true 6 pred 6	 #11 PASS true 6 pred 6	 #12 PASS true 8 pred 8	 #13 PASS true 3 pred 3	 #14 PASS true 2 pred 8	 #15 PASS true 8 pred 8
 #16 PASS true 4 pred 4	 #17 PASS true 7 pred 7	 #18 PASS true 9 pred 9	 #19 PASS true 5 pred 5	 #20 PASS true 3 pred 5	 #21 FAIL true 7 pred 7	 #22 PASS true 4 pred 4	 #23 PASS true 1 pred 1
 #24 PASS true 2 pred 2	 #25 PASS true 2 pred 1	 #26 PASS true 4 pred 4	 #27 PASS true 9 pred 9	 #28 PASS true 7 pred 7	 #29 PASS true 1 pred 1	 #30 PASS true 7 pred 7	 #31 PASS true 8 pred 8
 #32 PASS true 2 pred 2	 #33 PASS true 3 pred 3	 #34 PASS true 0 pred 0	 #35 PASS true 9 pred 9	 #36 PASS true 2 pred 6	 #37 PASS true 9 pred 9	 #38 PASS true 1 pred 1	 #39 PASS true 7 pred 7
 #40 PASS true 8 pred 8	 #41 PASS true 2 pred 2	 #42 PASS true 6 pred 6	 #43 PASS true 2 pred 3	 #44 PASS true 5 pred 5	 #45 PASS true 1 pred 1	 #46 FAIL true 2 pred 2	 #47 PASS true 6 pred 6
 #48 PASS true 0 pred 0	 #49 PASS true 0 pred 0	 #50 PASS true 8 pred 8	 #51 PASS true 8 pred 8	 #52 PASS true 2 pred 2	 #53 FAIL true 1 pred 1	 #54 PASS true 9 pred 9	 #55 PASS true 6 pred 6
 #56 PASS true 3 pred 3	 #57 PASS true 5 pred 0	 #58 FAIL true 8 pred 8	 #59 PASS true 6 pred 6	 #60 PASS true 9 pred 9	 #61 PASS true 2 pred 2	 #62 PASS true 7 pred 7	 #63 PASS true 0 pred 0

The four failures

#21: digit 7, smallest curvature -0.410 | #46: digit 2, smallest curvature -0.128 | #53: digit 1, smallest curvature -0.791 | #58: digit 8, smallest curvature -1.547

Why these four images failed

Each image had exactly one downhill direction out of 160 checked directions. The correction bars use a logarithmic scale because the gap is thousands-fold.



#21: true digit 7 -> predicted 7 | FAIL

Classifier confidence: 99.97% (correct)

Smallest curvature: -0.410227 | Downhill directions: 1 of 160

Gradient: 2341.3 before -> 16.38 after (142.9x reduction)

Reconstruction error: 0.006784 | MAP score: 525.56

The smallest possible repair was more than 4,102x larger than the allowed correction.



#46: true digit 2 -> predicted 2 | FAIL

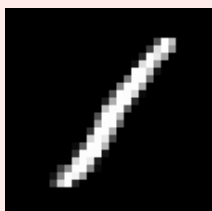
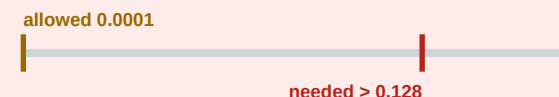
Classifier confidence: 100.00% (correct)

Smallest curvature: -0.128087 | Downhill directions: 1 of 160

Gradient: 2152.7 before -> 13.07 after (164.6x reduction)

Reconstruction error: 0.012103 | MAP score: 290.35

The smallest possible repair was more than 1,281x larger than the allowed correction.



#53: true digit 1 -> predicted 1 | FAIL

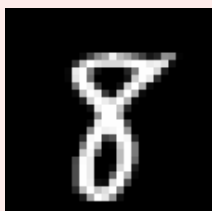
Classifier confidence: 99.98% (correct)

Smallest curvature: -0.791042 | Downhill directions: 1 of 160

Gradient: 2335.2 before -> 15.23 after (153.3x reduction)

Reconstruction error: 0.004822 | MAP score: 620.29

The smallest possible repair was more than 7,910x larger than the allowed correction.



#58: true digit 8 -> predicted 8 | FAIL

Classifier confidence: 91.92% (correct)

Smallest curvature: -1.547025 | Downhill directions: 1 of 160

Gradient: 2273.8 before -> 12.08 after (188.3x reduction)

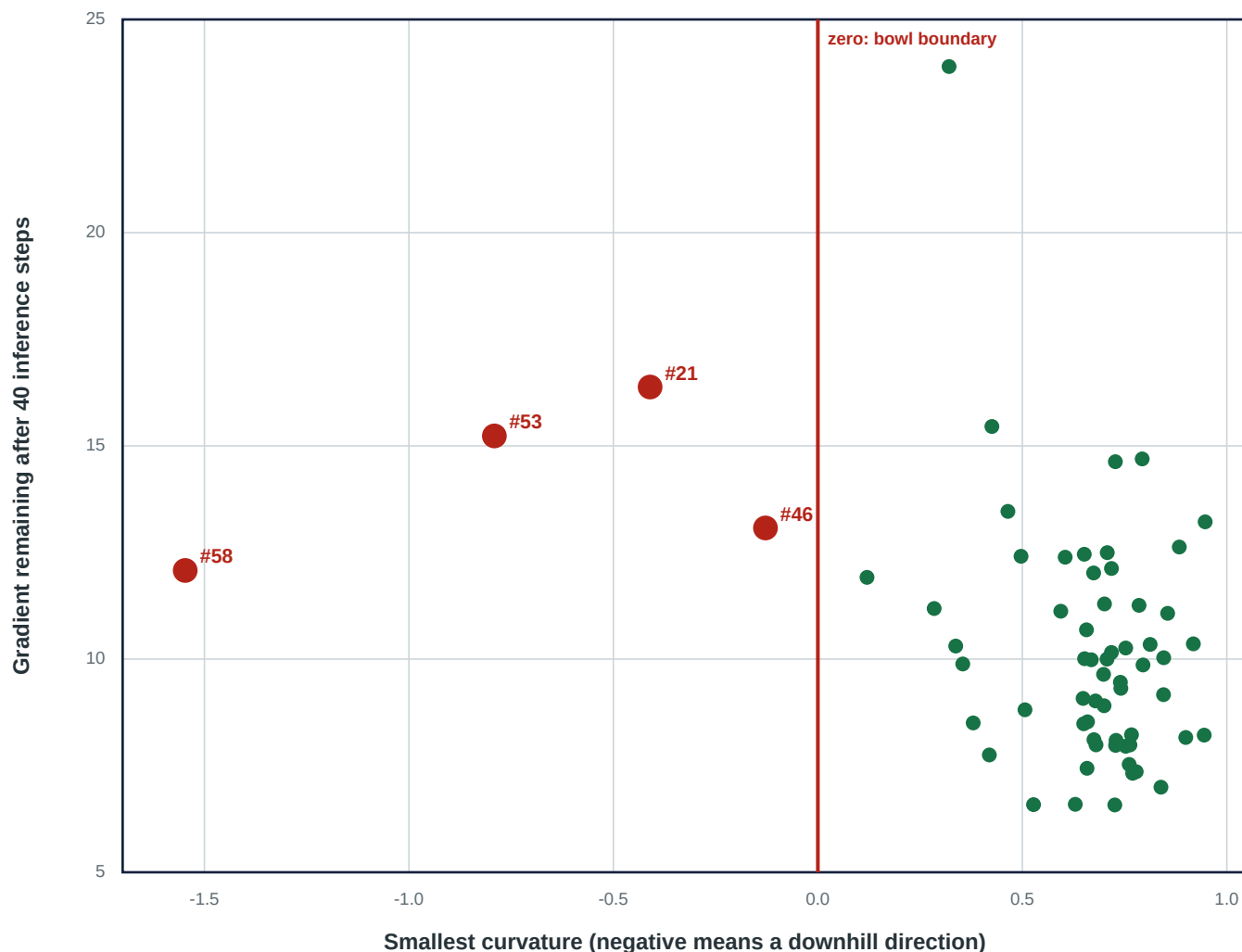
Reconstruction error: 0.007769 | MAP score: 477.26

The smallest possible repair was more than 15,470x larger than the allowed correction.



What separated passing images from failures

The horizontal axis is the weakest direction around the model's final internal state. Positive values mean every direction bends upward. Negative values mean at least one direction still goes downhill.



Passing images

All 60 passing images had positive minimum curvature, ranging from 0.120 to 0.947. Their final states were bowl-shaped in every checked direction.

Failing images

All four failures had exactly one negative direction. Their minimum curvatures were -0.128, -0.410, -0.791, and -1.547. These were not borderline round-off errors.

Recognition was not the cause

The classifier got all four failed digits right. Seven other images were classified incorrectly but still passed the curvature check.

Per-image measurements: #00 to #07

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#00 PASS



true 2 -> predicted 2

confidence 99.30%
classification correct

Source row	3577
Minimum curvature	+0.707665
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2251.5 -> 12.50 (180.2x)
Reconstruction error	0.008663
Residual score	-485.88
Complete MAP score	447.30
Score with 1e-8	504.83
Score with 1e-6	504.83
Score with 1e-4	504.82

#01 PASS



true 4 -> predicted 4

confidence 99.99%
classification correct

Source row	8925
Minimum curvature	+0.717924
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2220.8 -> 10.15 (218.7x)
Reconstruction error	0.007536
Residual score	-466.24
Complete MAP score	467.42
Score with 1e-8	526.98
Score with 1e-6	526.98
Score with 1e-4	526.98

#02 PASS



true 4 -> predicted 7

confidence 98.24%
classification wrong

Source row	1634
Minimum curvature	+0.717965
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2360.2 -> 12.12 (194.7x)
Reconstruction error	0.004040
Residual score	-291.30
Complete MAP score	641.01
Score with 1e-8	714.41
Score with 1e-6	714.41
Score with 1e-4	714.41

#03 PASS

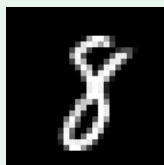


true 0 -> predicted 0

confidence 99.87%
classification correct

Source row	485
Minimum curvature	+0.761316
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2265.5 -> 7.53 (300.9x)
Reconstruction error	0.003965
Residual score	-303.99
Complete MAP score	628.53
Score with 1e-8	685.43
Score with 1e-6	685.43
Score with 1e-4	685.42

#04 PASS

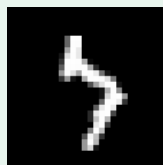


true 8 -> predicted 8

confidence 56.51%
classification correct

Source row	4753
Minimum curvature	+0.855339
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2342.5 -> 11.07 (211.6x)
Reconstruction error	0.005770
Residual score	-368.19
Complete MAP score	563.76
Score with 1e-8	621.18
Score with 1e-6	621.18
Score with 1e-4	621.18

#05 PASS



true 7 -> predicted 7

confidence 100.00%
classification correct

Source row	726
Minimum curvature	+0.752785
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2362.6 -> 10.26 (230.3x)
Reconstruction error	0.004101
Residual score	-305.63
Complete MAP score	626.61
Score with 1e-8	697.21
Score with 1e-6	697.20
Score with 1e-4	697.20

#06 PASS



true 7 -> predicted 7

confidence 100.00%
classification correct

Source row	9505
Minimum curvature	+0.425570
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2293.8 -> 15.45 (148.4x)
Reconstruction error	0.007077
Residual score	-423.54
Complete MAP score	509.70
Score with 1e-8	575.55
Score with 1e-6	575.55
Score with 1e-4	575.55

#07 PASS



true 0 -> predicted 0

confidence 99.73%
classification correct

Source row	3650
Minimum curvature	+0.769767
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2189.6 -> 7.32 (299.1x)
Reconstruction error	0.003773
Residual score	-300.95
Complete MAP score	632.02
Score with 1e-8	689.55
Score with 1e-6	689.55
Score with 1e-4	689.54

Per-image measurements: #08 to #15

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#08 PASS



true 3 -> predicted 3

confidence 99.79%
classification correct

Source row	4703
Minimum curvature	+0.604790
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2104.2 -> 12.39 (169.8x)
Reconstruction error	0.012585
Residual score	-676.80
Complete MAP score	258.80
Score with 1e-8	302.87
Score with 1e-6	302.87
Score with 1e-4	302.87

#09 PASS

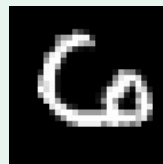


true 5 -> predicted 5

confidence 89.47%
classification correct

Source row	9428
Minimum curvature	+0.883828
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2315.2 -> 12.63 (183.3x)
Reconstruction error	0.005229
Residual score	-342.59
Complete MAP score	589.48
Score with 1e-8	650.38
Score with 1e-6	650.38
Score with 1e-4	650.37

#10 PASS



true 6 -> predicted 6

confidence 96.04%
classification correct

Source row	3939
Minimum curvature	+0.506465
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2055.4 -> 8.81 (233.3x)
Reconstruction error	0.007634
Residual score	-502.58
Complete MAP score	432.59
Score with 1e-8	477.77
Score with 1e-6	477.77
Score with 1e-4	477.77

#11 PASS



true 6 -> predicted 6

confidence 98.76%
classification correct

Source row	2286
Minimum curvature	+0.785355
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2166.9 -> 11.26 (192.4x)
Reconstruction error	0.008320
Residual score	-496.22
Complete MAP score	437.46
Score with 1e-8	487.97
Score with 1e-6	487.97
Score with 1e-4	487.96

#12 PASS



true 8 -> predicted 8

confidence 88.58%
classification correct

Source row	3492
Minimum curvature	+0.778487
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2269.3 -> 7.36 (308.4x)
Reconstruction error	0.003355
Residual score	-276.38
Complete MAP score	656.69
Score with 1e-8	716.01
Score with 1e-6	716.01
Score with 1e-4	716.00

#13 PASS

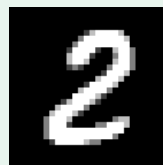


true 3 -> predicted 3

confidence 99.99%
classification correct

Source row	9022
Minimum curvature	+0.700893
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2030.0 -> 11.29 (179.8x)
Reconstruction error	0.011975
Residual score	-657.59
Complete MAP score	276.42
Score with 1e-8	310.54
Score with 1e-6	310.54
Score with 1e-4	310.54

#14 PASS



true 2 -> predicted 8

confidence 60.74%
classification wrong

Source row	5562
Minimum curvature	+0.726147
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2173.7 -> 6.58 (330.4x)
Reconstruction error	0.004033
Residual score	-327.61
Complete MAP score	605.81
Score with 1e-8	664.61
Score with 1e-6	664.61
Score with 1e-4	664.61

#15 PASS



true 8 -> predicted 8

confidence 99.95%
classification correct

Source row	6464
Minimum curvature	+0.766912
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2213.8 -> 8.22 (269.2x)
Reconstruction error	0.005223
Residual score	-376.20
Complete MAP score	557.55
Score with 1e-8	607.78
Score with 1e-6	607.78
Score with 1e-4	607.77

Per-image measurements: #16 to #23

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#16 PASS



true 4 -> predicted 4

confidence 81.52%
classification correct

Source row	1547
Minimum curvature	+0.496858
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2354.3 -> 12.41 (189.7x)
Reconstruction error	0.004445
Residual score	-300.22
Complete MAP score	631.69
Score with 1e-8	697.75
Score with 1e-6	697.75
Score with 1e-4	697.74

#17 PASS

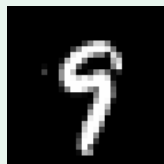


true 7 -> predicted 7

confidence 70.63%
classification correct

Source row	2888
Minimum curvature	+0.680189
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2363.2 -> 7.99 (295.8x)
Reconstruction error	0.002089
Residual score	-207.73
Complete MAP score	724.10
Score with 1e-8	804.11
Score with 1e-6	804.11
Score with 1e-4	804.10

#18 PASS



true 9 -> predicted 9

confidence 99.51%
classification correct

Source row	9483
Minimum curvature	+0.899654
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2388.7 -> 8.16 (292.7x)
Reconstruction error	0.002581
Residual score	-220.72
Complete MAP score	710.46
Score with 1e-8	781.83
Score with 1e-6	781.83
Score with 1e-4	781.82

#19 PASS



true 5 -> predicted 5

confidence 99.99%
classification correct

Source row	9685
Minimum curvature	+0.650126
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2254.0 -> 8.48 (265.7x)
Reconstruction error	0.004135
Residual score	-315.06
Complete MAP score	618.39
Score with 1e-8	681.24
Score with 1e-6	681.24
Score with 1e-4	681.23

#20 PASS

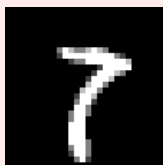


true 3 -> predicted 5

confidence 54.07%
classification wrong

Source row	5358
Minimum curvature	+0.763132
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2242.1 -> 7.99 (280.7x)
Reconstruction error	0.003602
Residual score	-296.71
Complete MAP score	636.66
Score with 1e-8	696.63
Score with 1e-6	696.63
Score with 1e-4	696.62

#21 FAIL



true 7 -> predicted 7

confidence 99.97%
classification correct

Source row	3055
Minimum curvature	-0.410227
Downhill directions	1 of 160
Shift needed	0.410227
Largest allowed shift	0.000100
Gradient	2341.3 -> 16.38 (142.9x)
Reconstruction error	0.006784
Residual score	-406.67
Complete MAP score	525.56
Score with 1e-8	FAILED
Score with 1e-6	FAILED
Score with 1e-4	FAILED

#22 PASS

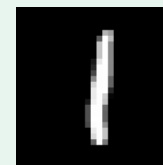


true 4 -> predicted 4

confidence 99.97%
classification correct

Source row	9615
Minimum curvature	+0.648603
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2081.9 -> 9.08 (229.3x)
Reconstruction error	0.009818
Residual score	-581.65
Complete MAP score	353.60
Score with 1e-8	400.22
Score with 1e-6	400.22
Score with 1e-4	400.22

#23 PASS



true 1 -> predicted 1

confidence 99.98%
classification correct

Source row	3648
Minimum curvature	+0.944735
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2424.8 -> 8.22 (295.1x)
Reconstruction error	0.001482
Residual score	-164.31
Complete MAP score	766.40
Score with 1e-8	857.36
Score with 1e-6	857.36
Score with 1e-4	857.35

Per-image measurements: #24 to #31

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#24 PASS



true 2 -> predicted 2

confidence 99.75%
classification correct

Source row	4506
Minimum curvature	+0.678951
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2157.5 -> 9.02 (239.3x)
Reconstruction error	0.006886
Residual score	-440.08
Complete MAP score	493.39
Score with 1e-8	539.69
Score with 1e-6	539.69
Score with 1e-4	539.69

#25 PASS

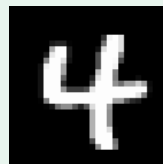


true 2 -> predicted 1

confidence 86.58%
classification wrong

Source row	3436
Minimum curvature	+0.320835
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2307.8 -> 23.90 (96.6x)
Reconstruction error	0.008272
Residual score	-460.53
Complete MAP score	472.03
Score with 1e-8	533.88
Score with 1e-6	533.88
Score with 1e-4	533.88

#26 PASS

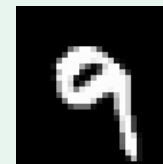


true 4 -> predicted 4

confidence 99.99%
classification correct

Source row	9064
Minimum curvature	+0.707068
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2122.1 -> 10.00 (212.2x)
Reconstruction error	0.007448
Residual score	-483.22
Complete MAP score	451.47
Score with 1e-8	500.41
Score with 1e-6	500.41
Score with 1e-4	500.41

#27 PASS

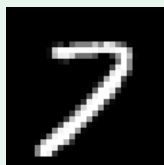


true 9 -> predicted 9

confidence 99.90%
classification correct

Source row	4240
Minimum curvature	+0.652219
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2206.0 -> 10.01 (220.4x)
Reconstruction error	0.007529
Residual score	-466.23
Complete MAP score	467.87
Score with 1e-8	528.13
Score with 1e-6	528.13
Score with 1e-4	528.13

#28 PASS

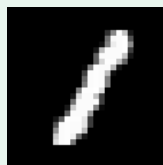


true 7 -> predicted 7

confidence 82.87%
classification correct

Source row	7504
Minimum curvature	+0.918189
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2259.7 -> 10.36 (218.2x)
Reconstruction error	0.007879
Residual score	-449.48
Complete MAP score	483.17
Score with 1e-8	543.97
Score with 1e-6	543.97
Score with 1e-4	543.97

#29 PASS



true 1 -> predicted 1

confidence 99.87%
classification correct

Source row	8360
Minimum curvature	+0.419304
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2306.4 -> 7.75 (297.5x)
Reconstruction error	0.002391
Residual score	-216.71
Complete MAP score	715.95
Score with 1e-8	797.15
Score with 1e-6	797.15
Score with 1e-4	797.14

#30 PASS



true 7 -> predicted 7

confidence 100.00%
classification correct

Source row	4539
Minimum curvature	+0.337134
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2164.2 -> 10.31 (210.0x)
Reconstruction error	0.007674
Residual score	-477.10
Complete MAP score	457.72
Score with 1e-8	515.66
Score with 1e-6	515.66
Score with 1e-4	515.65

#31 PASS



true 8 -> predicted 8

confidence 98.25%
classification correct

Source row	3559
Minimum curvature	+0.464739
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2225.7 -> 13.46 (165.3x)
Reconstruction error	0.010050
Residual score	-564.43
Complete MAP score	369.80
Score with 1e-8	408.10
Score with 1e-6	408.10
Score with 1e-4	408.09

Per-image measurements: #32 to #39

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#32 PASS



true 2 -> predicted 2

confidence 99.99%
classification correct

Source row	4713
Minimum curvature	+0.120189
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	1980.7 -> 11.92 (166.2x)
Reconstruction error	0.013924
Residual score	-752.59
Complete MAP score	182.91
Score with 1e-8	221.87
Score with 1e-6	221.87
Score with 1e-4	221.86

#33 PASS



true 3 -> predicted 3

confidence 89.53%
classification correct

Source row	2008
Minimum curvature	+0.838985
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2308.7 -> 7.00 (330.0x)
Reconstruction error	0.002995
Residual score	-257.64
Complete MAP score	674.62
Score with 1e-8	741.62
Score with 1e-6	741.62
Score with 1e-4	741.62

#34 PASS



true 0 -> predicted 0

confidence 99.97%
classification correct

Source row	7702
Minimum curvature	+0.658191
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2203.8 -> 7.44 (296.3x)
Reconstruction error	0.004330
Residual score	-329.45
Complete MAP score	604.00
Score with 1e-8	657.17
Score with 1e-6	657.17
Score with 1e-4	657.17

#35 PASS



true 9 -> predicted 9

confidence 100.00%
classification correct

Source row	5977
Minimum curvature	+0.698639
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2056.2 -> 9.64 (213.3x)
Reconstruction error	0.011200
Residual score	-631.76
Complete MAP score	302.92
Score with 1e-8	353.64
Score with 1e-6	353.64
Score with 1e-4	353.64

#36 PASS



true 2 -> predicted 6

confidence 99.19%
classification wrong

Source row	6352
Minimum curvature	+0.651465
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	1972.7 -> 12.46 (158.3x)
Reconstruction error	0.010068
Residual score	-607.03
Complete MAP score	328.31
Score with 1e-8	376.55
Score with 1e-6	376.55
Score with 1e-4	376.55

#37 PASS

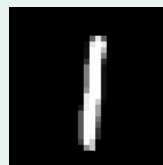


true 9 -> predicted 9

confidence 99.54%
classification correct

Source row	108
Minimum curvature	+0.740912
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2317.2 -> 9.31 (248.8x)
Reconstruction error	0.003428
Residual score	-274.89
Complete MAP score	657.79
Score with 1e-8	726.67
Score with 1e-6	726.67
Score with 1e-4	726.67

#38 PASS

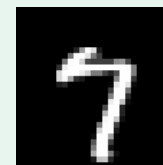


true 1 -> predicted 1

confidence 99.80%
classification correct

Source row	1867
Minimum curvature	+0.845248
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2430.4 -> 9.17 (265.2x)
Reconstruction error	0.001599
Residual score	-171.75
Complete MAP score	759.00
Score with 1e-8	852.05
Score with 1e-6	852.05
Score with 1e-4	852.04

#39 PASS



true 7 -> predicted 7

confidence 99.89%
classification correct

Source row	415
Minimum curvature	+0.354375
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2324.6 -> 9.88 (235.2x)
Reconstruction error	0.004581
Residual score	-323.18
Complete MAP score	609.88
Score with 1e-8	681.71
Score with 1e-6	681.71
Score with 1e-4	681.71

Per-image measurements: #40 to #47

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#40 PASS



true 8 -> predicted 8

confidence 94.51%
classification correct

Source row	1675
Minimum curvature	+0.728104
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2193.1 -> 7.98 (275.0x)
Reconstruction error	0.004881
Residual score	-359.52
Complete MAP score	574.49
Score with 1e-8	627.40
Score with 1e-6	627.40
Score with 1e-4	627.40

#41 PASS



true 2 -> predicted 2

confidence 99.55%
classification correct

Source row	258
Minimum curvature	+0.629476
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2109.1 -> 6.59 (319.9x)
Reconstruction error	0.004595
Residual score	-358.23
Complete MAP score	576.36
Score with 1e-8	633.83
Score with 1e-6	633.83
Score with 1e-4	633.83

#42 PASS



true 6 -> predicted 6

confidence 100.00%
classification correct

Source row	8133
Minimum curvature	+0.675135
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2145.4 -> 8.11 (264.6x)
Reconstruction error	0.006890
Residual score	-436.49
Complete MAP score	497.69
Score with 1e-8	552.31
Score with 1e-6	552.31
Score with 1e-4	552.31

#43 PASS



true 2 -> predicted 3

confidence 98.96%
classification wrong

Source row	3681
Minimum curvature	+0.812513
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2054.4 -> 10.34 (198.6x)
Reconstruction error	0.008957
Residual score	-534.39
Complete MAP score	399.51
Score with 1e-8	441.55
Score with 1e-6	441.55
Score with 1e-4	441.55

#44 PASS

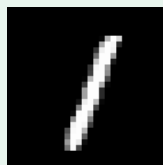


true 5 -> predicted 5

confidence 99.92%
classification correct

Source row	6638
Minimum curvature	+0.793098
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2290.2 -> 14.69 (155.9x)
Reconstruction error	0.008578
Residual score	-467.68
Complete MAP score	464.47
Score with 1e-8	520.82
Score with 1e-6	520.82
Score with 1e-4	520.81

#45 PASS



true 1 -> predicted 1

confidence 95.80%
classification correct

Source row	7708
Minimum curvature	+0.739724
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2396.9 -> 9.46 (253.5x)
Reconstruction error	0.002011
Residual score	-185.45
Complete MAP score	745.65
Score with 1e-8	834.27
Score with 1e-6	834.27
Score with 1e-4	834.27

#46 FAIL



true 2 -> predicted 2

confidence 100.00%
classification correct

Source row	8295
Minimum curvature	-0.128087
Downhill directions	1 of 160
Shift needed	0.128087
Largest allowed shift	0.000100
Gradient	2152.7 -> 13.07 (164.6x)
Reconstruction error	0.012103
Residual score	-643.85
Complete MAP score	290.35
Score with 1e-8	FAILED
Score with 1e-6	FAILED
Score with 1e-4	FAILED

#47 PASS



true 6 -> predicted 6

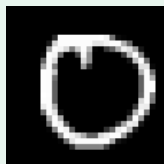
confidence 100.00%
classification correct

Source row	5263
Minimum curvature	+0.668680
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2214.2 -> 9.98 (221.8x)
Reconstruction error	0.006731
Residual score	-417.16
Complete MAP score	516.46
Score with 1e-8	574.78
Score with 1e-6	574.78
Score with 1e-4	574.78

Per-image measurements: #48 to #55

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#48 PASS

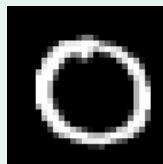


true 0 -> predicted 0

confidence 98.09%
classification correct

Source row	305
Minimum curvature	+0.594036
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	1953.2 -> 11.12 (175.6x)
Reconstruction error	0.014234
Residual score	-766.24
Complete MAP score	169.18
Score with 1e-8	207.39
Score with 1e-6	207.39
Score with 1e-4	207.39

#49 PASS



true 0 -> predicted 0

confidence 100.00%
classification correct

Source row	3849
Minimum curvature	+0.379864
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	1975.1 -> 8.50 (232.3x)
Reconstruction error	0.006197
Residual score	-442.09
Complete MAP score	492.98
Score with 1e-8	540.56
Score with 1e-6	540.56
Score with 1e-4	540.55

#50 PASS



true 8 -> predicted 8

confidence 99.86%
classification correct

Source row	5184
Minimum curvature	+0.727514
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2281.3 -> 14.63 (155.9x)
Reconstruction error	0.007455
Residual score	-446.66
Complete MAP score	486.42
Score with 1e-8	533.29
Score with 1e-6	533.29
Score with 1e-4	533.28

#51 PASS



true 8 -> predicted 8

confidence 99.53%
classification correct

Source row	5049
Minimum curvature	+0.674310
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2262.0 -> 12.02 (188.2x)
Reconstruction error	0.006355
Residual score	-400.16
Complete MAP score	532.54
Score with 1e-8	586.93
Score with 1e-6	586.93
Score with 1e-4	586.93

#52 PASS

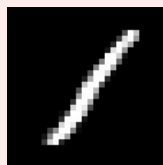


true 2 -> predicted 2

confidence 99.01%
classification correct

Source row	2085
Minimum curvature	+0.659358
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2213.1 -> 8.53 (259.5x)
Reconstruction error	0.005922
Residual score	-382.13
Complete MAP score	551.26
Score with 1e-8	611.33
Score with 1e-6	611.33
Score with 1e-4	611.33

#53 FAIL



true 1 -> predicted 1

confidence 99.98%
classification correct

Source row	9774
Minimum curvature	-0.791042
Downhill directions	1 of 160
Shift needed	0.791042
Largest allowed shift	0.000100
Gradient	2335.2 -> 15.23 (153.3x)
Reconstruction error	0.004822
Residual score	-311.98
Complete MAP score	620.29
Score with 1e-8	FAILED
Score with 1e-6	FAILED
Score with 1e-4	FAILED

#54 PASS



true 9 -> predicted 9

confidence 99.96%
classification correct

Source row	3693
Minimum curvature	+0.845688
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2347.0 -> 10.03 (234.0x)
Reconstruction error	0.003742
Residual score	-288.07
Complete MAP score	643.62
Score with 1e-8	715.12
Score with 1e-6	715.12
Score with 1e-4	715.12

#55 PASS



true 6 -> predicted 6

confidence 99.76%
classification correct

Source row	3688
Minimum curvature	+0.947092
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2366.0 -> 13.22 (179.0x)
Reconstruction error	0.005642
Residual score	-352.86
Complete MAP score	578.67
Score with 1e-8	644.50
Score with 1e-6	644.50
Score with 1e-4	644.50

Per-image measurements: #56 to #63

Minimum curvature is the decisive value: positive means a bowl in all 160 directions; negative means at least one downhill direction. FAILED means the final probability approximation could not be computed.

#56 PASS

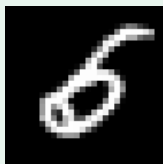


true 3 -> predicted 3

confidence 99.91%
classification correct

Source row	8842
Minimum curvature	+0.729215
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2112.2 -> 8.09 (261.0x)
Reconstruction error	0.005810
Residual score	-398.47
Complete MAP score	535.40
Score with 1e-8	585.05
Score with 1e-6	585.05
Score with 1e-4	585.05

#57 PASS

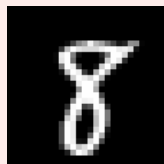


true 5 -> predicted 0

confidence 58.75%
classification wrong

Source row	9729
Minimum curvature	+0.656845
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2164.7 -> 10.69 (202.6x)
Reconstruction error	0.010898
Residual score	-591.08
Complete MAP score	342.85
Score with 1e-8	391.17
Score with 1e-6	391.17
Score with 1e-4	391.16

#58 FAIL



true 8 -> predicted 8

confidence 91.92%
classification correct

Source row	3704
Minimum curvature	-1.547025
Downhill directions	1 of 160
Shift needed	1.547025
Largest allowed shift	0.000100
Gradient	2273.8 -> 12.08 (188.3x)
Reconstruction error	0.007769
Residual score	-455.79
Complete MAP score	477.26
Score with 1e-8	FAILED
Score with 1e-6	FAILED
Score with 1e-4	FAILED

#59 PASS

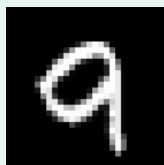


true 6 -> predicted 6

confidence 100.00%
classification correct

Source row	8024
Minimum curvature	+0.527373
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2181.2 -> 6.59 (331.2x)
Reconstruction error	0.003610
Residual score	-310.37
Complete MAP score	623.77
Score with 1e-8	689.92
Score with 1e-6	689.92
Score with 1e-4	689.92

#60 PASS



true 9 -> predicted 9

confidence 97.51%
classification correct

Source row	1501
Minimum curvature	+0.795078
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2235.8 -> 9.86 (226.7x)
Reconstruction error	0.006333
Residual score	-413.61
Complete MAP score	519.56
Score with 1e-8	575.08
Score with 1e-6	575.08
Score with 1e-4	575.07

#61 PASS



true 2 -> predicted 2

confidence 67.61%
classification correct

Source row	3909
Minimum curvature	+0.284393
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2231.8 -> 11.19 (199.5x)
Reconstruction error	0.006455
Residual score	-400.58
Complete MAP score	532.24
Score with 1e-8	592.60
Score with 1e-6	592.60
Score with 1e-4	592.59

#62 PASS

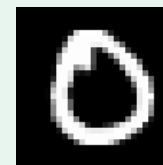


true 7 -> predicted 7

confidence 98.45%
classification correct

Source row	8749
Minimum curvature	+0.752816
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2321.1 -> 7.96 (291.8x)
Reconstruction error	0.003011
Residual score	-257.37
Complete MAP score	675.95
Score with 1e-8	749.58
Score with 1e-6	749.58
Score with 1e-4	749.58

#63 PASS



true 0 -> predicted 0

confidence 100.00%
classification correct

Source row	5684
Minimum curvature	+0.699848
Downhill directions	0 of 160
Shift needed	0.000000
Largest allowed shift	0.000100
Gradient	2002.6 -> 8.91 (224.9x)
Reconstruction error	0.005676
Residual score	-398.43
Complete MAP score	535.35
Score with 1e-8	583.83
Score with 1e-6	583.83
Score with 1e-4	583.82

What this result means

What succeeded

The implementation's exact probability formula passed its analytic check. All eight one-node training choices passed the basic learning requirements. The selected model learned recognizable digits, and it produced valid curvature-based scores for 60 of the 64 audit images.

What failed

After 40 internal adjustment steps, four images were still at saddle points rather than bowl bottoms. Each saddle had one downhill direction. The largest allowed correction, 0.0001, was between 1,281 and 15,470 times smaller than the correction needed merely to reach zero curvature.

What we cannot conclude

No temporal nodes were built, so this run did not test whether residual, MAP, or curvature-corrected scores route images to the right node. It did not test merging. It does not show that predictive-coding routing succeeds or fails.

The clearest next experiment

Keep the trained model and these exact 64 images. Run internal inference longer, or stop only after the remaining gradient falls below a fixed absolute limit. Then check whether images 21, 46, 53, and 58 move from saddles to bowl bottoms without changing the score formula. Only after all 64 produce valid bowls should the temporal routing experiment run.

Why not simply use a much larger correction?

A correction as large as 1.55 would change every model's score, not merely repair a rounding error. It could hide incomplete inference and distort comparisons between nodes. That would be a new experiment, not a harmless numerical fix.